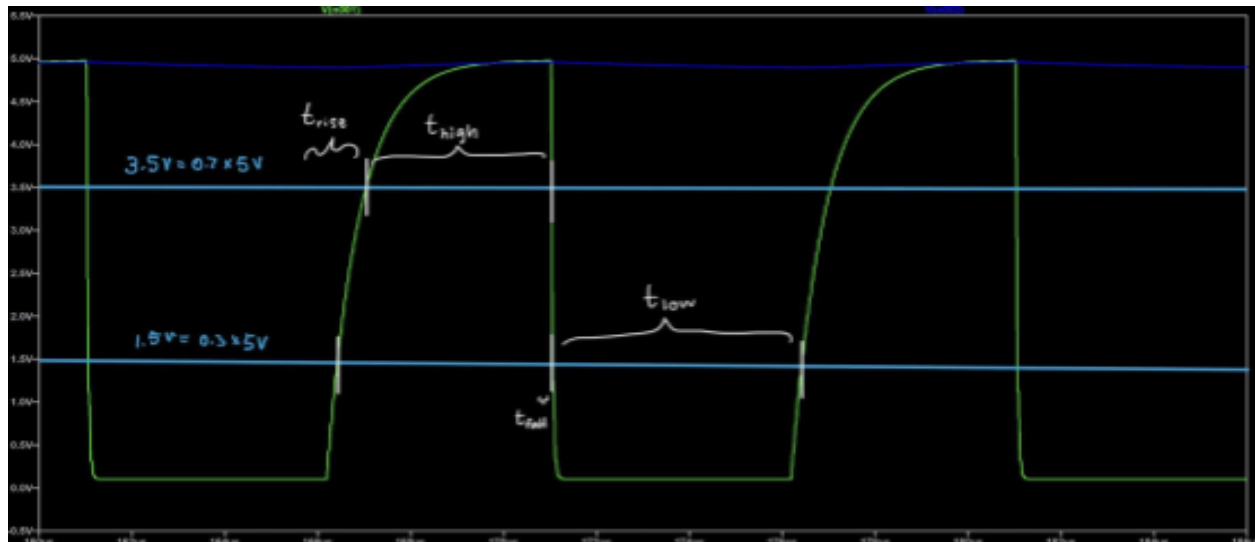


Introduction

This report determines suitable pull-up resistor values for the I2C bus. Too high a resistance and the line rises too slowly; too low and sink current limits are violated. These two constraints define a theoretical valid resistance window. A range of values around this window are then tested in LTSpice simulation against the six I2C specifications shown below:

$$V_{\text{low}} = 0.3 \cdot V_{\text{cc}} \mid V_{\text{high}} = 0.7 \cdot V_{\text{cc}} \mid t_{\text{rise}} \mid t_{\text{high}} \mid t_{\text{fall}} \mid t_{\text{low}}$$

Visual of Specification Values:



e.g 100KHz, 8-device channel waveform

Table 11. Characteristics of the SDA and SCL bus lines for Standard, Fast, and Fast-mode Plus I²C-bus devices

All values referred to $V_{\text{H(min)}}$ (0.7V_{DD}) and $V_{\text{L(max)}}$ (0.3V_{DD}) levels (see Table 10).

Symbol	Parameter	Conditions	Standard-mode		Fast-mode		Fast-mode Plus		Unit
			Min	Max	Min	Max	Min	Max	
f_{SCL}	SCL clock frequency		0	100	0	400	0	1000	kHz
$t_{\text{HD,STA}}$	hold time (repeated) START condition	After this period, the first clock pulse is generated.	4.0	-	0.6	-	0.26	-	μs
t_{LOW}	LOW period of the SCL clock		4.7	-	1.3	-	0.5	-	μs
t_{HIGH}	HIGH period of the SCL clock		4.0	-	0.6	-	0.26	-	μs
$t_{\text{SU,STA}}$	set-up time for a repeated START condition		4.7	-	0.6	-	0.26	-	μs
$t_{\text{HD,DAT}}$	data hold time ^[1]	CBUS compatible controllers (see Remark in Section 4.1)	5.0	-	-	-	-	-	μs
		I ² C-bus devices	0 ^[2]	- ^[3]	0 ^[2]	- ^[3]	0	-	μs
$t_{\text{SU,DAT}}$	data set-up time		250	-	100 ^[4]	-	50	-	ns
t_r	rise time of both SDA and SCL signals		-	1000	20	300	-	120	ns
t_f	fall time of both SDA and SCL signals ^{[2] [5] [6] [7]}		-	300	$20 \times (V_{\text{DD}} / 5.5 \text{ V})$	300	$20 \times (V_{\text{DD}} / 5.5 \text{ V})$ ^[8]	120 ^[7]	ns

<https://www.nxp.com/docs/en/user-guide/UM10204.pdf> pg44

Limit to I^2C Communication Speed

The maximum operating frequency of an I²C bus is reached when the design constraints on the pull-up resistors conflict—specifically, when the *minimum* pull-up resistance required for reliable logic-level operation becomes larger than the *maximum* pull-up resistance allowed to meet timing requirements.

The **minimum pull-up resistance** R_{min} is set by the need to ensure that devices can pull the line low without exceeding their maximum sink current. It is given by:

$$R_{min} = \frac{V_{DD} - V_{OL(max)}}{I_{OL(max)}}$$

This ensures that when a device drives the line low, the resulting voltage does not exceed the maximum allowable low-level voltage $V_{OL(max)}$.

The **maximum pull-up resistance** R_{max} is determined by the requirement that the signal rise time is fast enough to meet the bus timing specifications. Since the rising edge is governed by the RC time constant of the pull-up resistor and bus capacitance, it is approximated by:

$$R_{max} = \frac{t_{rise}}{0.8473 * C_b}$$

where t_{rise} is the maximum allowable rise time and C_b is the total bus capacitance.

As the desired communication frequency increases, the allowable rise time t_{rise} decreases, which in turn lowers R_{max} . The system reaches its maximum feasible speed when $R_{min} > R_{max}$, at which point no resistor value can simultaneously satisfy both the current-driving and timing constraints.

I2C Bus Values Base Assumptions

Characteristic	Value	Known/Assumption
8-device channel Bus Capacitance(C_b)	200pF	Assumption
4-device channel Bus Capacitance(C_b)	160pF	Assumption
Bus High Voltage(V_{cc})	5.0V	Known
Sensors' Sink Current(I_{OL})	3mA	Known
Sensors' Voltage Output Low (V_{OL})	0.4V	Known

- Based on the [MCP9808 Datasheet](#)(pg4), $V_{OL} = 0.4V$ and $I_{OL} = 3mA$. It adds a capacitance of 5pF to the bus line.
- Based on the [INA260 Datasheet](#)pg(7) $V_{OL} = 0.4V$ and $I_{OL} = 3mA$. It adds a capacitance of 3pF to the bus line.
- Based on the [TCA9548A Datasheet](#)pg(6) $V_{OL} = 0.4V$ and $I_{OL} = 3mA$. It adds a capacitance of ~28pF-36pF to the bus line. Assume a worst case of ~36pF.
- Based on this [Wire Capacitance Calculator](#), using $r = 0.01in$ (24AWG wire), $h = 12in$, $L=1m$, $\epsilon=1$, the capacitance of each wire is ~7.1pF.
- Based on this [Wire Inductance Calculator](#), using $d=0.02in$ (24AWG wire), $L=36in$, the inductance of each wire is ~1.5uH.

$$C_b \approx C_{bus\ sensor} + C_{wire} + C_{mux} + C_{extra, just\ in\ case}$$

8-device Channel Temperature Sensor:

$$C_b \approx 8(5pF) + 8(7.1pF) + 36pF + 70pF = 202.8pF$$

4-device Channel Current Sensor:

$$C_b \approx 4(3pF) + 4(7.1pF) + 36pF + 70pF = 146.4pF$$

Theoretical Max/Min Pullup Resistor Values

$$R(\min) = \frac{V_{cc} - V_{OL}}{I_{OL(\max)}} = \frac{5 - 0.4}{3 \times 10^{-3}} \Omega = 1533 \Omega$$

Standard Mode(100KHz): $(t_{rise(\max)} = 1000ns)$

8-device channel:

$$R(\max) = \frac{t_{rise}}{0.8473 * C_b} = \frac{(1000 * 10^{-9})}{0.8473 * (200 * 10^{-12})} = 5.9k\Omega$$

4-device channel:

$$R(\max) = \frac{t_{rise}}{0.8473 * C_b} = \frac{(1000 * 10^{-9})}{0.8473 * (160 * 10^{-12})} = 7.38k\Omega$$

Fast-Mode(400KHz): $(t_{rise(\max)} = 300ns)$

8-device channel:

$$R(\max) = \frac{t_{rise}}{0.8473 * C_b} = \frac{(300 * 10^{-9})}{0.8473 * (200 * 10^{-12})} = 1.77k\Omega$$

4-device channel:

$$R(\max) = \frac{t_{rise}}{0.8473 * C_b} = \frac{(300 * 10^{-9})}{0.8473 * (160 * 10^{-12})} = 2.2k\Omega$$

Fast-Mode Plus(1MHz): $(t_{rise(\max)} = 120ns)$

8-device channel:

$$R(\max) = \frac{t_{rise}}{0.8473 * C_b} = \frac{(120 * 10^{-9})}{0.8473 * (200 * 10^{-12})} = 708\Omega$$

4-device channel:

$$R(\max) = \frac{t_{rise}}{0.8473 * C_b} = \frac{(120 * 10^{-9})}{0.8473 * (160 * 10^{-12})} = 885\Omega$$

This Desmos model can be used to find the average power dissipation based of the value of pullup resistors, bus capacitance, frequency of communication, and duty cycle:

[Power Dissipation Graph](#)

Default Values: (V=5V, D=0.4, f=100kHz, C=200pF)

Note: $P(R) = 2 * \text{Expression}$. Expression is multiplied by 2 to account for cumulative power dissipation of both SCL and SDA lines.

Conclusion:

The I²C communication breaks down when operating in Fast-Mode Plus(1MHz) because $708,885\Omega > 1533\Omega$. Fast Mode(400kHz) is the fastest reliable communication speed.

LTspice Simulation of I²C Bus

Introduction:

This report documents an LTspice simulation of an I2C bus network used to verify signal integrity and timing compliance for a system in which a microcontroller communicates outward to sixteen peripheral devices through a single TCA9548A 8-channel multiplexer. The devices under simulation are the MCP9808 temperature sensor and the INA260 current and power monitor.

Because manufacturer-provided SPICE models are not available for any of the integrated circuits in this system, all components are represented using behavioral approximations constructed from their published electrical characteristics.

The system operates at a supply voltage of 5 V. Pull-up resistors will be selected to balance rise time performance against sink current requirements under the total capacitive load presented by the network.

Circuit Model:

The I2C bus is modeled as a passive electrical network in LTSpice, scoped to the physical layer only. No active device logic, protocol sequencing, or acknowledgment behavior is included.

SCL line — Modeled as a voltage-controlled switch to ground driven by a pulse source, replicating the open-drain behavior of I2C. The switch pulls SCL low when closed and releases it to the pull-up when open. The PULSE settings model the specific frequency the line is modeled at. SDA is modeled identically.

Mux channels — The TCA9548A is represented as three independently controlled switches, one per active channel, each driven by a non-overlapping pulse source.

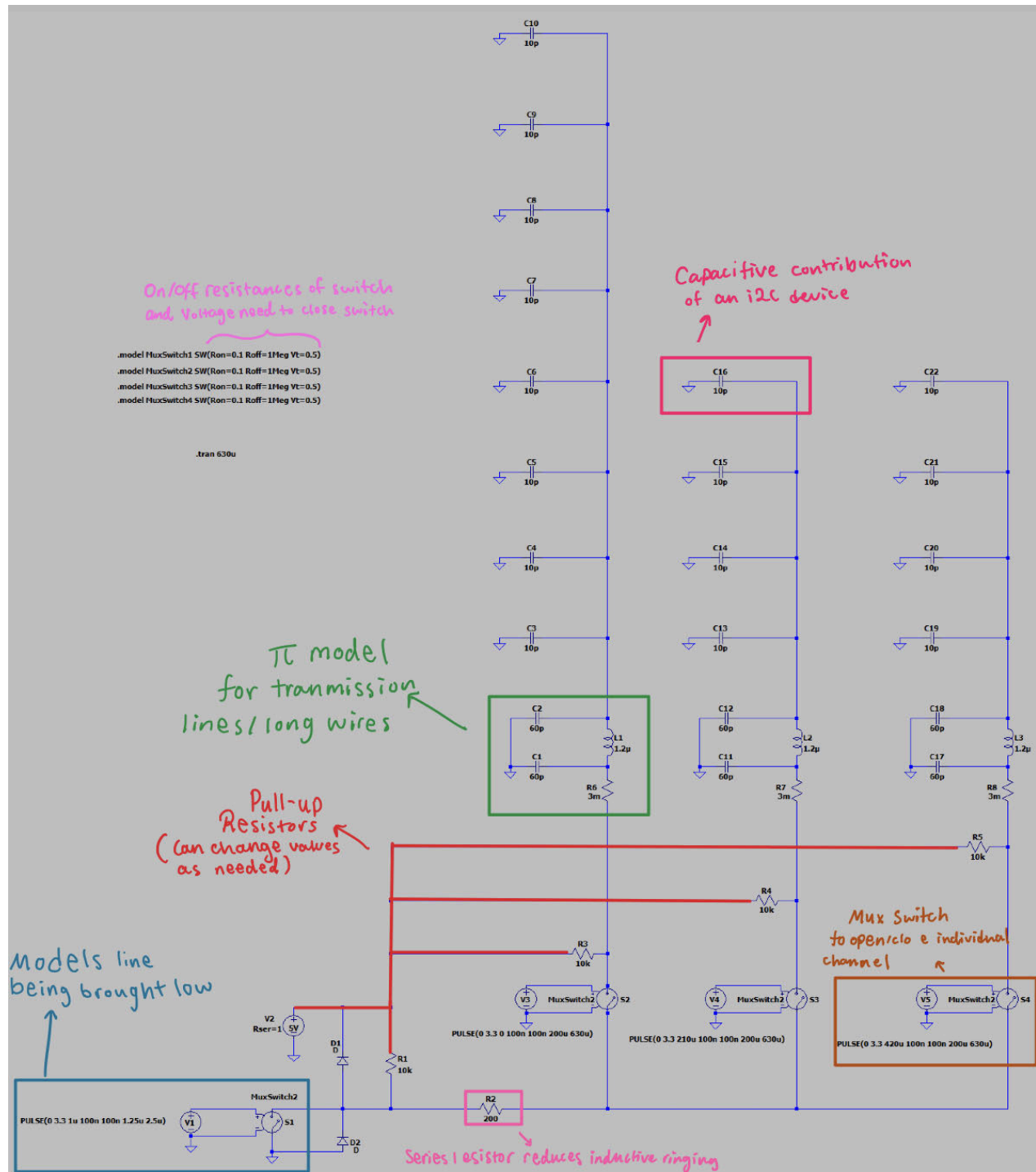
Sensors — Each device pin is modeled as a 10pF capacitor to ground. This is appropriate for a physical layer simulation since an I2C input pin presents primarily a capacitive load during passive reception.

Pull-up resistors — The main upstream line and each downstream mux channel carry a dedicated pull-up resistor to 5V.

Wire model — Each wire run is represented as a pi-section lumped network — shunt capacitance at each end with series resistance and inductance between them — realistically

capturing the distributed parasitics of 1–3ft wire runs without requiring a full transmission line model.

Series damping resistor — A resistor placed in series with the main bus line suppresses inductive ringing on fast falling edges by damping the LC resonance formed by wire inductance



$\text{PULSE}(V_{\text{low}}, V_{\text{high}}, \text{Delay}, \text{Rise Time}, T_{\text{on}}, \text{Period})$

V2,V3,V4 are set to PULSE(0, 3.3, X, 100n, 200u, 630u) with delay(X) being different for each so no two channels are opened at the same time.

Data & Findings

The following simulations have fixed values $R1 = 10\text{k}\Omega$, $R2 = 200\Omega$. Total pull-up resistance can be found on this [Calculator](#).

Data Collection for simulations can be found on this [Spreadsheet](#).

Standard Mode(100KHz):

The following settings will be used to model the limit of 100kHz communication speed:

- V1: PULSE(0, 3.3, 1u, 100n, 5u, 10u)
“0-3.3V, delay 1u before it starts pulsing, it takes 100n to change from 0-3.3V, it is 3.3V for 5u with a period of 10u

Finding:

- 8-device channel Highest resistance that all specifications were met: $R2 = 14\text{k}\Omega$ (total pullup resistance of $\sim 5.87\text{k}\Omega$)
- 4-device channel Highest resistance that all specifications were met: $R3 = 20\text{k}\Omega$ (total pullup resistance of $\sim 7.2\text{k}\Omega$)

Fast-Mode(400KHz):

The following settings will be used to model the limit of 400kHz communication speed:

- V1: PULSE(0, 3.3, 1u, 100n, 1.25u, 2.5u)
“0-3.3V, delay 1u before it starts pulsing, it takes 100n to change from 0-3.3V, it is 3.3V for 1.25u with a period of 2.5u

Finding:

- 8-device channel Highest resistance that all specifications were met: $R2 = 2.1\text{k}\Omega$ (total pullup resistance of $\sim 1.87\text{k}\Omega$)

- 4-device channel Highest resistance that all specifications were met: $R_3 = 2.5\text{k}\Omega$ (total pullup resistance of $\sim 2.1\text{k}\Omega$)

Conclusion

The I2C bus will use a $10\text{k}\Omega$ on the mcu side of the multiplexer for each line. A 200Ω resistor will be in series with the bus to reduce ringing. Operating the 8-device channel's pullup resistor at $2\text{k}\Omega$ (total resistance of $1.8\text{k}\Omega$), and the 4-device channel's pullup at $2.2\text{k}\Omega$ (total resistance of $1.94\text{k}\Omega$) allow for both 100kHz and 400kHz I2C communication specification to be met in simulation.

Main Sources:

Information for Resistor Max/Min: [TI paper](#)

Specification Terms Meaning: [Analog Devices page](#):

Specification tested in Simulation: [I2C Specification](#)